

## OPTI-CROP PROJECT DELIVERABLE

# Project Executable Files & Installation Guide

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## 1. System Requirements

- **Python:** Python 3.9 or higher (Python 3.11 recommended).
- **RAM:** Minimum 2GB.
- **Disk Space:** 500MB free.

## 2. Local Installation Steps

### Step 1: Clone the Repository

Clone the Opti-Crop code repository locally.

```
git clone https://github.com/AmbicaSairam/Opti-Crop.git
cd Opti-Crop
```

### Step 2: Set Up Virtual Environment

Create a virtual environment to manage dependencies:

```
python -m venv venv
# Activate on Windows:
.\venv\Scripts\activate
# Activate on Linux/macOS:
source venv/bin/activate
```

### Step 3: Install Required Libraries

```
pip install -r "5. Project Development Phase/requirements.txt"
```

### Step 4: Initialize the SQLite Database

```
python "5. Project Development Phase/database.py"
```

### Step 5: Execute Model Training & Assets Generation

```
python "5. Project Development Phase/train_model.py"
```

### Step 6: Start the Application Server

```
python "5. Project Development Phase/app.py"
```

The application will launch on **http://127.0.0.1:5000/**.